

Week 01. Part 2. Descriptive statistics

Univariate Data. Central tendency. Box plot.

Eduard Bukin

eduard.bukin@agrار.uni-giessen.de

Justus Liebig University Giessen

Institute of Agricultural Policy and Market Research

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The Univariate Data

- Data that consists of a **single variable**.
- **Univariate data analysis**: descriptive analysis of a single variable.
- What statistical measurements are used to describe a variable?
 - Central Tendency.
 - Variance
 - Symmetry.

The Central Tendency

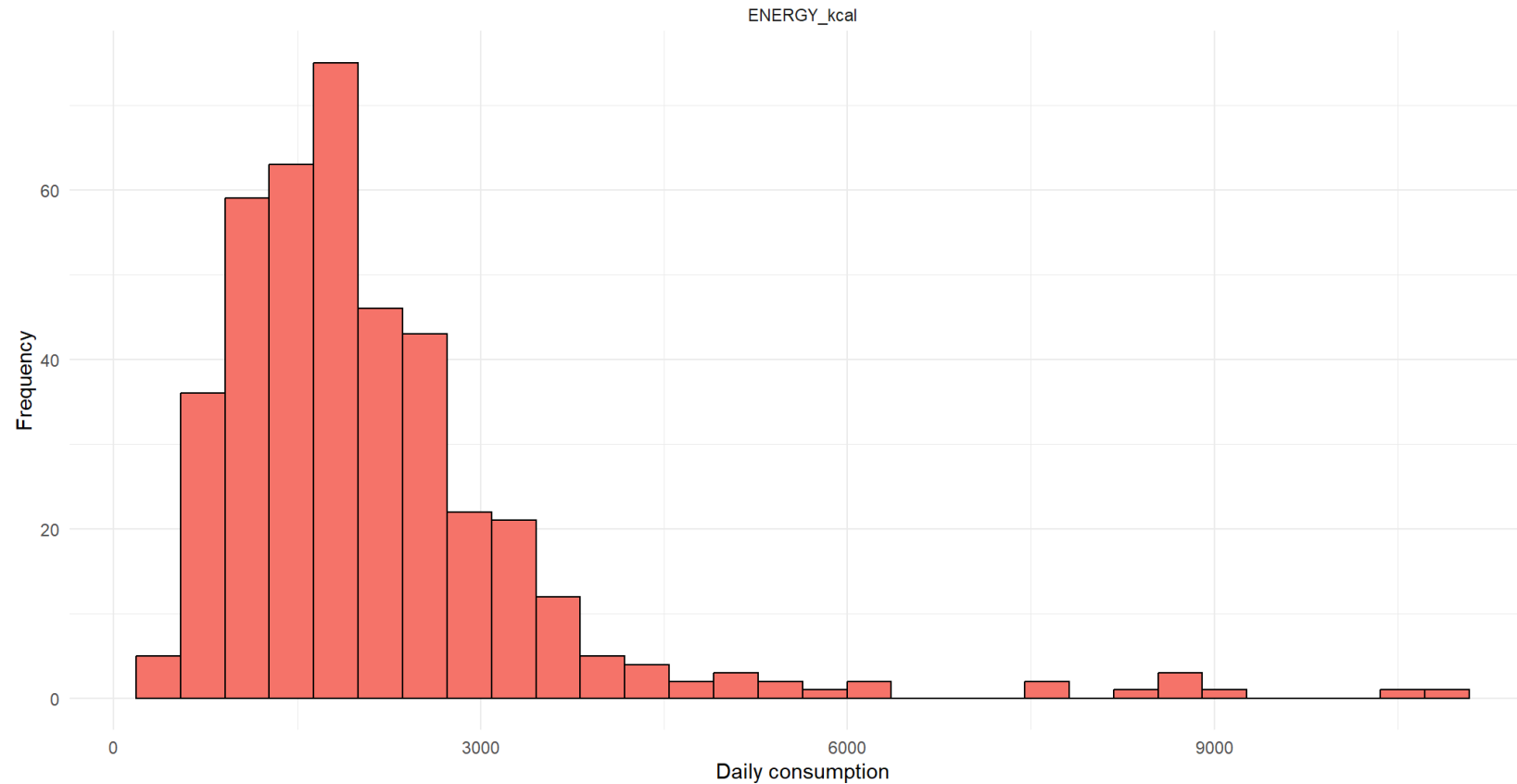
- It is a single value that describes where values $\{x_1, x_2, x_3, \dots, x_n\}$ of a variable x cluster around
- Most commonly measures are:
 - Median; Mode; Arithmetic Mean; Weighted Mean;
 - Other measures of the center;

Individuals' consumption of nutrients in the DR Congo

- We use the survey of 395 individuals from the Democratic Republic of Congo.
- It covers daily consumption of key nutrients and energy intake by population:
- Key variables are
 - **ENERGY_kcal** - energy consumption in kilo calories per day;
 - **PROTEIN_g** - protein consumption in grams;
 - **FAT_g** - fat consumption in grams;
 - **HEIGHT** and **WEIGHT** respondents height and weight in cm and kg respectively;
 - **SEX** - dummy variable that is 1 when respondent is a female and 0 if it is a male;
- Data source: “FAO/WHO GIFT | Global Individual Food” survey data [available here](#).

Describe a distribution

The DR Congo. Individuals' daily energy consumption

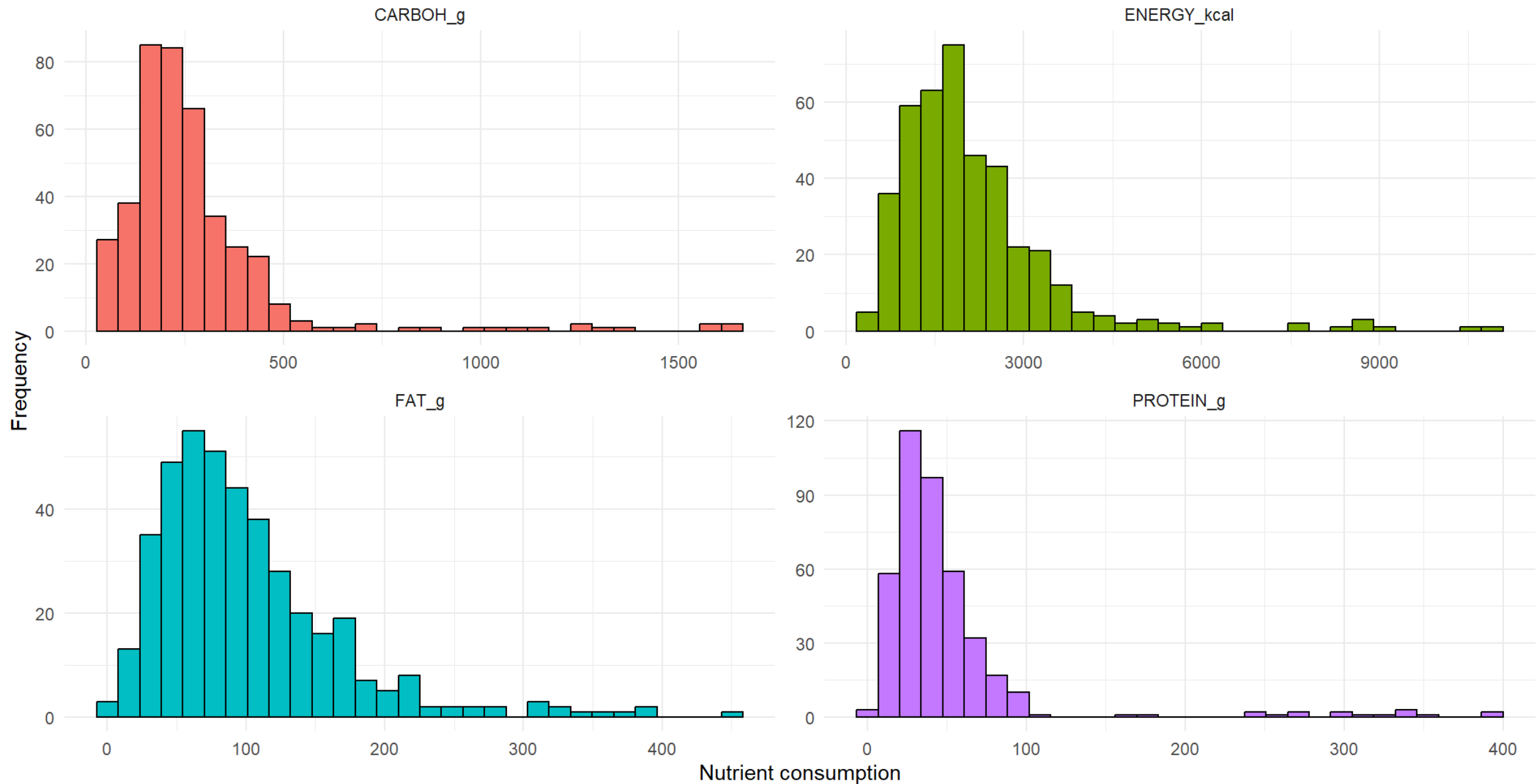


What are the:

- Central tendency
- Variance
- Symmetry of the distribution

At home: describe the rest of distributions

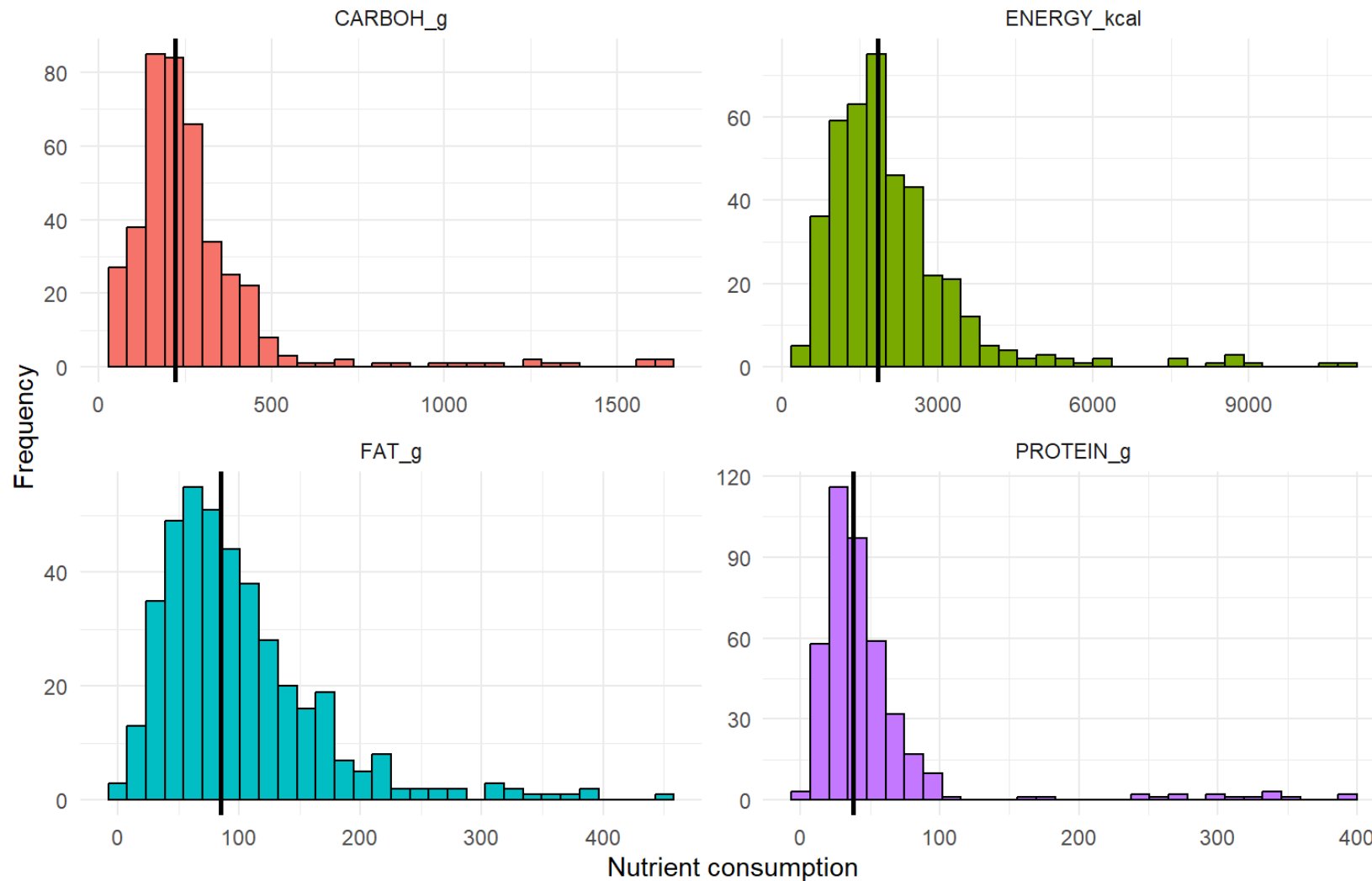
The DR Congo. Individuals' daily nutrients consumption



Central Tendency: the Median

The DR Congo. Individuals' daily nutrients consumption

Central tendency | Median



- Median is the value of the observation in the center
 - after 50% of observations and before the remaining 50%.
- Less susceptible to the extreme and influential observations.

The Arithmetic Mean (The Mean)

! Important

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N},$$

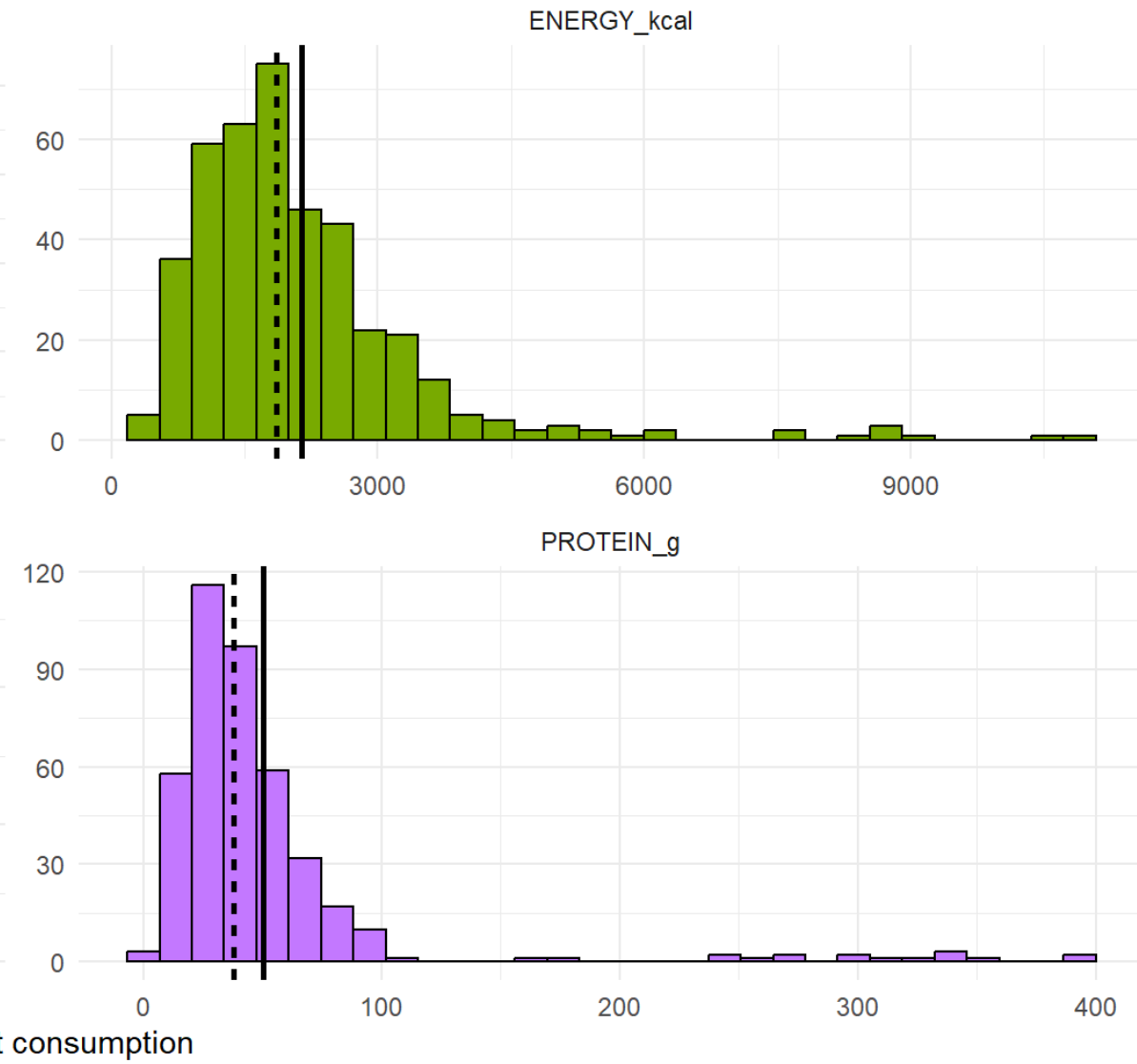
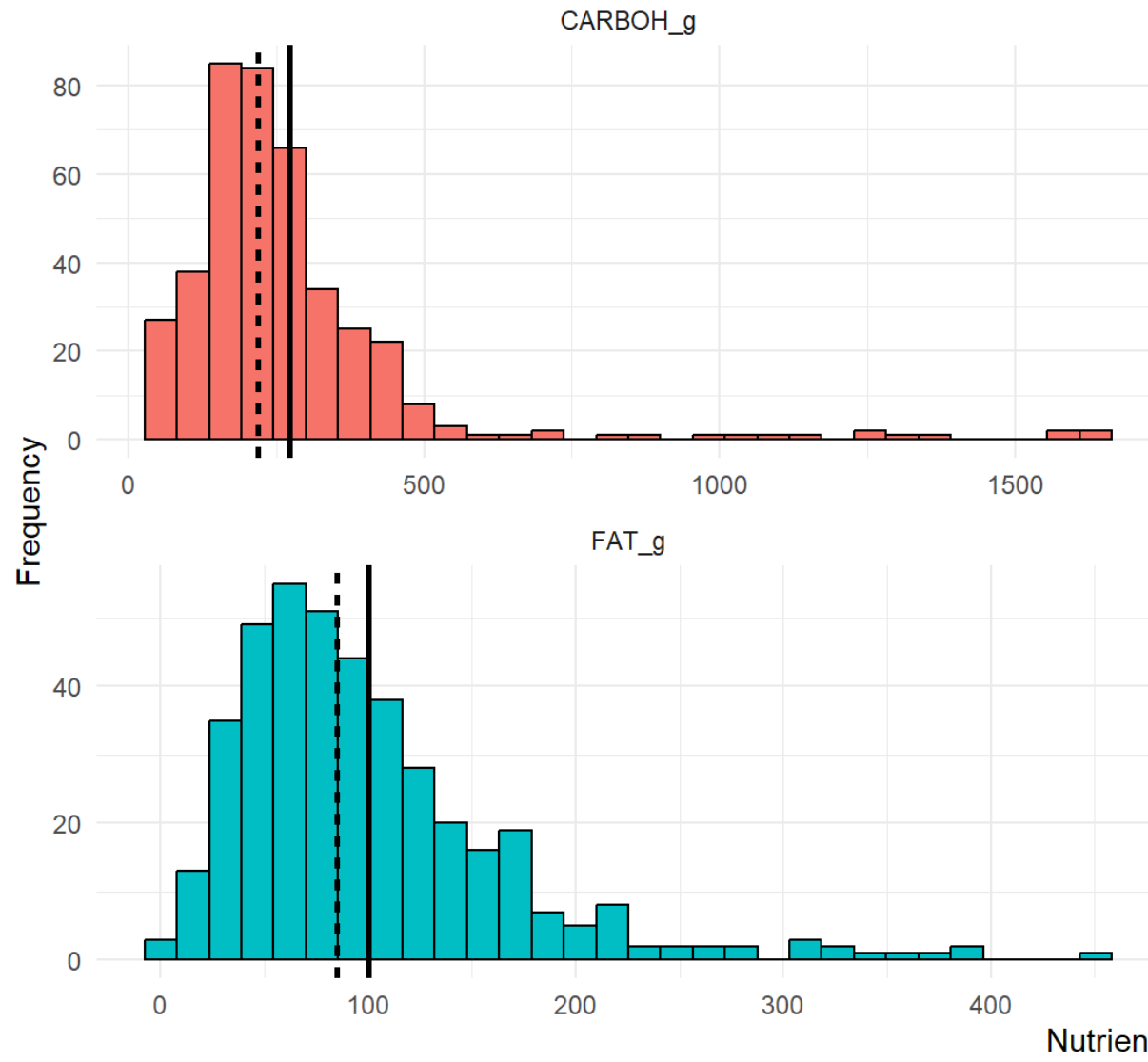
where,

- $\bar{x}$ is the central value of x , x could take any values.
 - N is the total number of observations.
-
- The Arithmetic Mean is affected by the outliers and may be misleading if the distribution is skewed.

The Mean (1)

The DR Congo. Individuals' daily nutrients consumption

Central tendency | Mean : Median



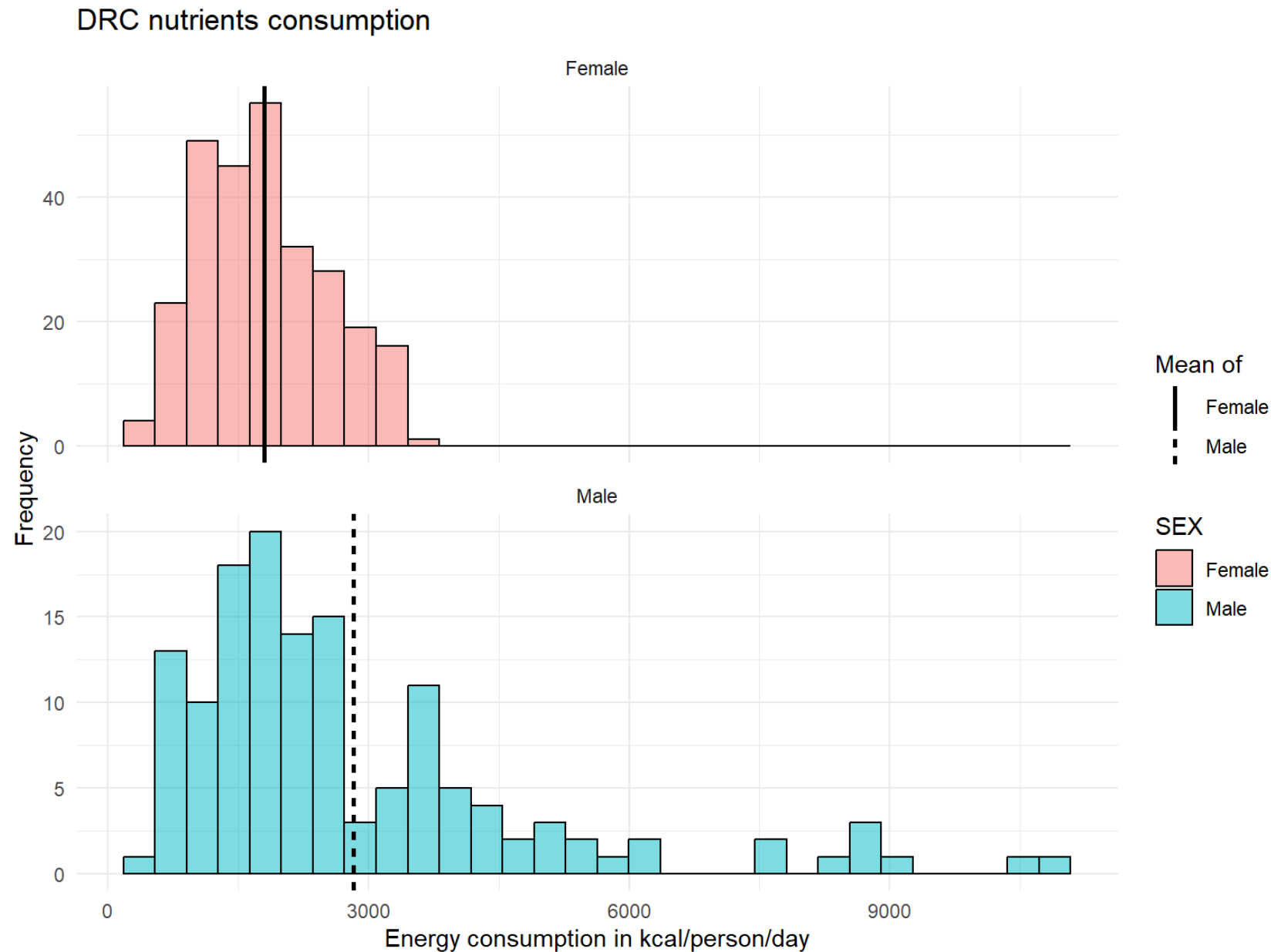
The Mean (2)

Summary statistics

Variable	Mean	Median
CARBOH_g	272.35	219.65
ENERGY_kcal	2146.48	1855.00
FAT_g	100.10	84.90
PROTEIN_g	50.13	37.90
SEX	0.72	1.00

- How to interpret the means?
- How to interpret the medians?
- Why means are different from medians?
- What does the **average of SEX** variable means?
- Why **median of SEX** variable is equal to 1?

Extreme/influential observations/outlier



Arithmetic mean is very sensitive to the extreme observations, outliers, and non-symmetric distribution.

Population may contain extreme values, however, such outliers appearing in a small sample, may substantially bias our estimates.

Dealing with extreme observations

1. Document them:

- Describe, where, why and how many extreme observations were found.
- Describe why such extreme observation could occur.

2. Analyze reasons behind extreme observations:

- Measurement error? Is this the true incidence in the population or an entry error? Is this a systematic entry error?
- Sampling error? Is this observation prevalent in the population or is this a random sampling error? Is it systematic?

3. Keep, adjust statistical method, and generalization level:

- Use “robust” measures (median/IQR) to describe data
- Consider data “[winsorizing](#)”

4. Remove some only in the exceptional cases!

The Weighted Average (The Weighted Mean)

! Important

$$\bar{x} = \frac{\sum_{i=1}^N x_i w_i}{\sum_{i=1}^N w_i},$$

where,

- $\bar{x}$ is the **estimate** of the central value of x , x could take any values.
 - N is the total number of observations.
 - x is the weight that each observation has in the data.
-
- the arithmetic average is essentially a weighted average, where the weight of each observations is equal to 1.

The Weighted Average example

Let us measure the level of extreme poverty in the World.

- poverty is the state of livelihood when by income or consumption individual is living below the poverty line;
- it measures:
 - the % of people that by their income (or consumption) live below a **poverty line**,
 - poverty line **per person per day** is country-specific or international in the international currency (e.g. [Purchasing Poverty Parity - PPP](#))
- extreme poverty is the poverty level given extremely low poverty line of 2.15 US\$ PPP per day.
- Data: World Bank [World Development Indicators](#);

The Data

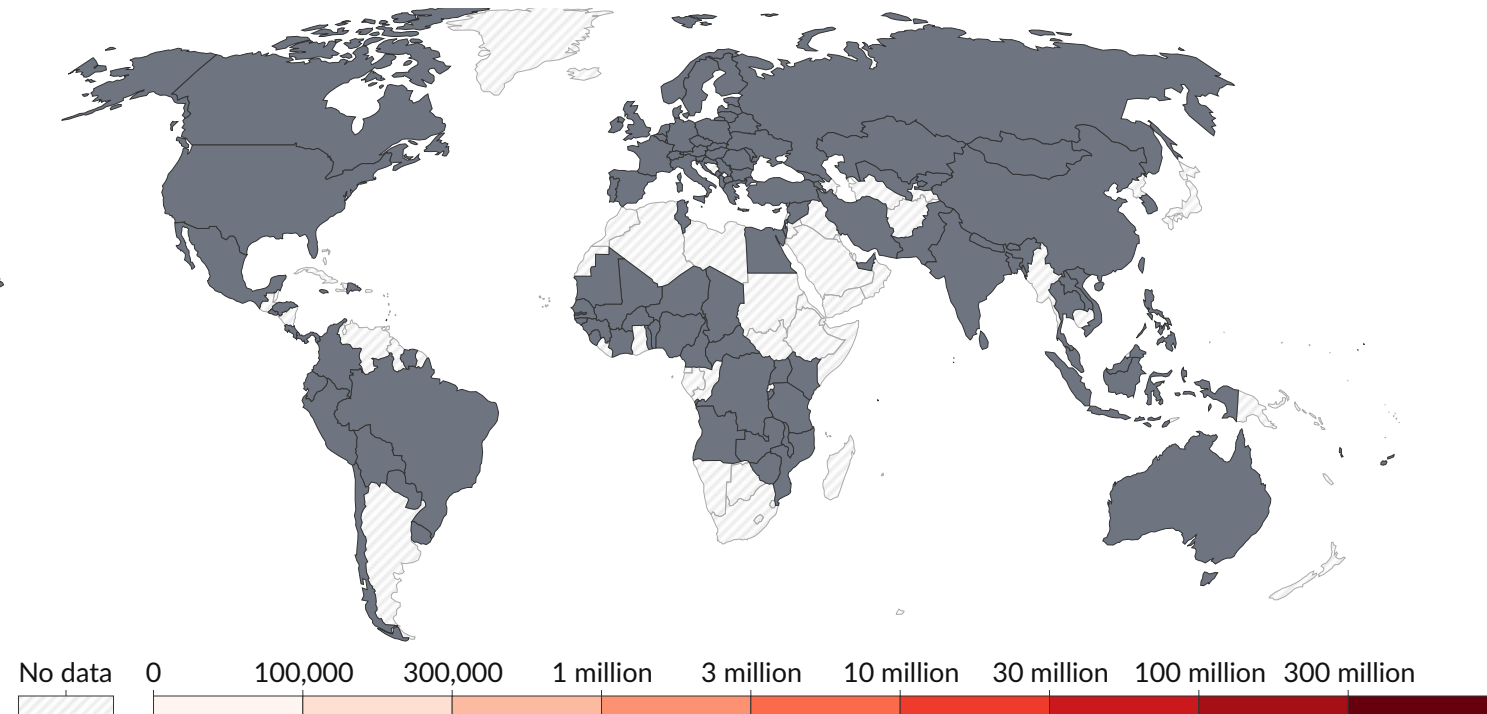
Number of people living in extreme poverty, 2023

Extreme poverty is defined as living below the International Poverty Line of \$2.15 per day. This data is adjusted for inflation and for differences in the cost of living between countries.

Our World
in Data

Table Map Chart

World



► Play time-lapse 1963

2023

Data source: World Bank Poverty and Inequality Platform (2024) – [Learn more about this data](#)

Note: This data is expressed in international-\$ at 2017 prices. Depending on the country and year, it relates to income measured after taxes and benefits, or to consumption, per capita.

OurWorldinData.org/poverty | CC BY



Explore the data →

Source: [Global Extreme Poverty](#). See also: [Poverty and inequality platform](#).
Understanding poverty and [Macro Poverty Outlook](#).

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Average extreme poverty in the world (1)

First 12 observations of the country-specific data

Country	Extreme poverty	Population
Bangladesh	14.3	166.3
Botswana	14.5	2.4
Nigeria	39.1	211.4
Fiji	1.8	0.9
Egypt, Arab Rep.	3.8	104.3
Moldova	0.0	2.6
Tajikistan	4.1	9.7
Mozambique	63.7	32.2
Micronesia, Fed. Sts.	15.4	0.1
Bolivia	4.4	11.8
United Arab Emirates	0.0	10.0
Finland	0.0	5.5

How to measure the extreme poverty in the world?

Mean	Median
12.56467	2

- Is any of those the right measure of poverty?
- Can we use any of those?

-  **Important**
Neither of those is correct because every country has a different size.

Average extreme poverty in the world (2)

1% extreme poverty rate in China
(1.4B people):

- implies 14 million people living in poverty.

49% poverty in Turkmenistan (6.1 M people):

- is 3 million people living in poverty.

- The arithmetic mean of these two is:

- $(1 \% + 49 \%) / 2 = 25 \%$;

- but, 25% of their total population is $0.25 * 1.406B = 352M$..

- Arithmetic mean implies that there is 351.5 M people living in poverty, but the sum of the number of poor people is $14M + 3M = 17M$ people.

Average extreme poverty in the world (3)

- Arithmetic means, **without accounting for the size/weight** is an inappropriate measure of average poverty.
- We must weight country-specific poverty level by the country's population.
 - $(1\% * 1.4B + 49\% * 6.1 M) / (1.4B + 6.1 M) = 1.2 \%$
- Weighting are essential in:
 - Analysis of the representative Individuals/Household surveys.
 - Macro-level data aggregation.

At home: Country sizes matters

Source: [The map we need if we want to think about how global living conditions are changing](#)



At home: Country sizes matters



At home: Exam problems on the central tendency

- When the mean may not be the best measure of the central tendency in our data?
- What measures of the central tendency we most often apply in social sciences?
- The estimated arithmetic mean of the household size in our survey of 22 observations is 7.5 person, while the median size is 4.
 - Which measure of the central tendency would you use to describe the average family size?
 - Why do you suggest using this measure?
 - What is the total number of individuals (not households) that is represented in our survey?

At home: a complex question on central tendency

Note

A representative household survey was collected from the population of all households in the country. Data contains many variables on various socioeconomic measures of the household along with the variable “Population weights”.

- What do those “Population weights” mean?
- How using the weighted statistics by “Population weights” may change estimates of the central tendency?
- What difference the weighted and non-weighted mean makes for the conclusions and generalization of the estimates of the central tendency?

At home: Calculate question

Extreme poverty	Population
9.2	27.1
18.3	30.5
NA	NA
1.5	0.8
3.4	2.3
0.0	43.8

Calculate:

- arithmetic average of **Extreme poverty** and
- average of **Extreme poverty** weighted by the **Population**:

Arithmetic average	Weighted average
6.5	7.8

The Box Plot

- Summaries one numeric variables on a single plot
- Presents key statistics about the distribution and central tendency:
 - Median;
 - upper (Q3) and lower (Q1) [quartiles](#);
 - IQR - [interquartile range](#), where 50% of observations are;
 - Whiskers and outliers
- At home, check: [THE BOXPLOT AND ITS PITFALLS](#)
- At home, check: [Wiki: the box plot](#)

The Box Plot

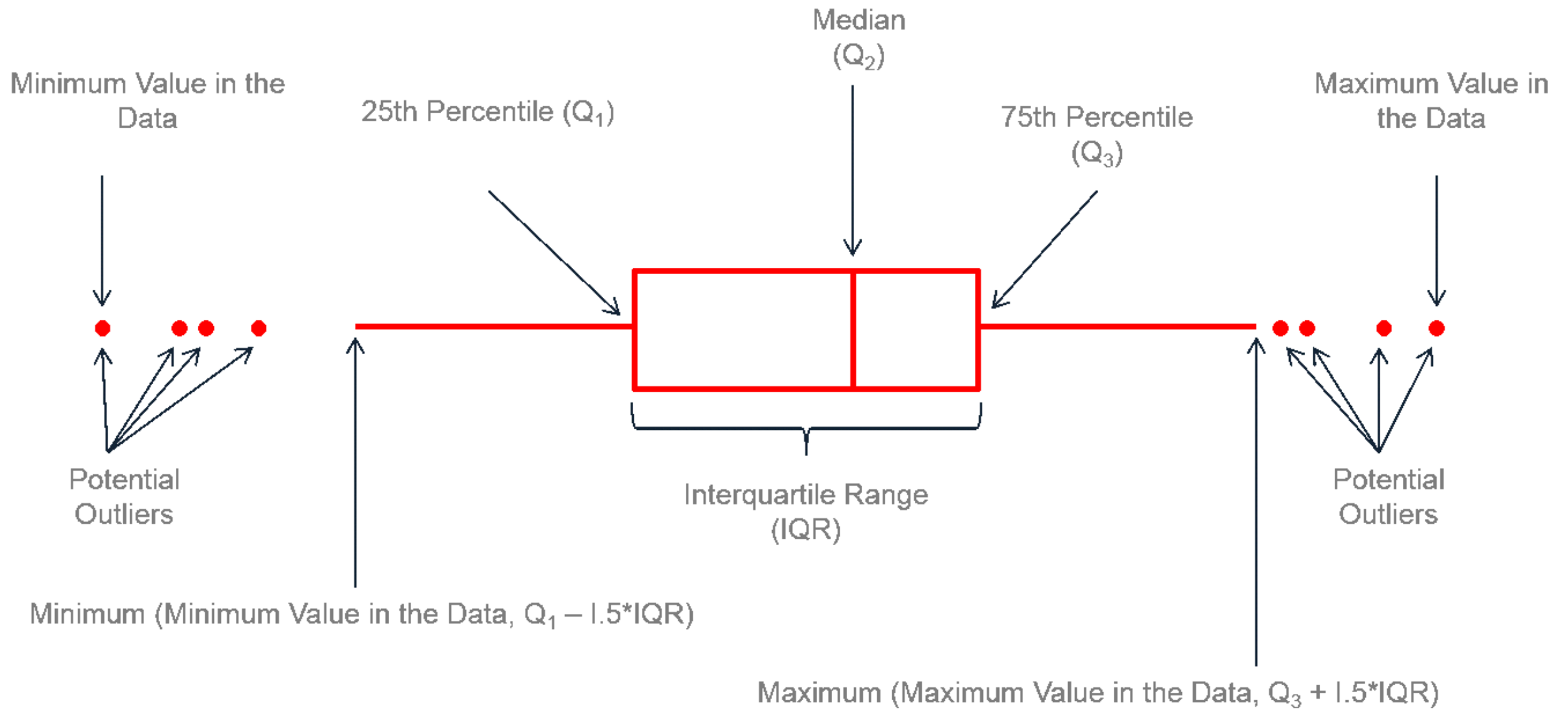
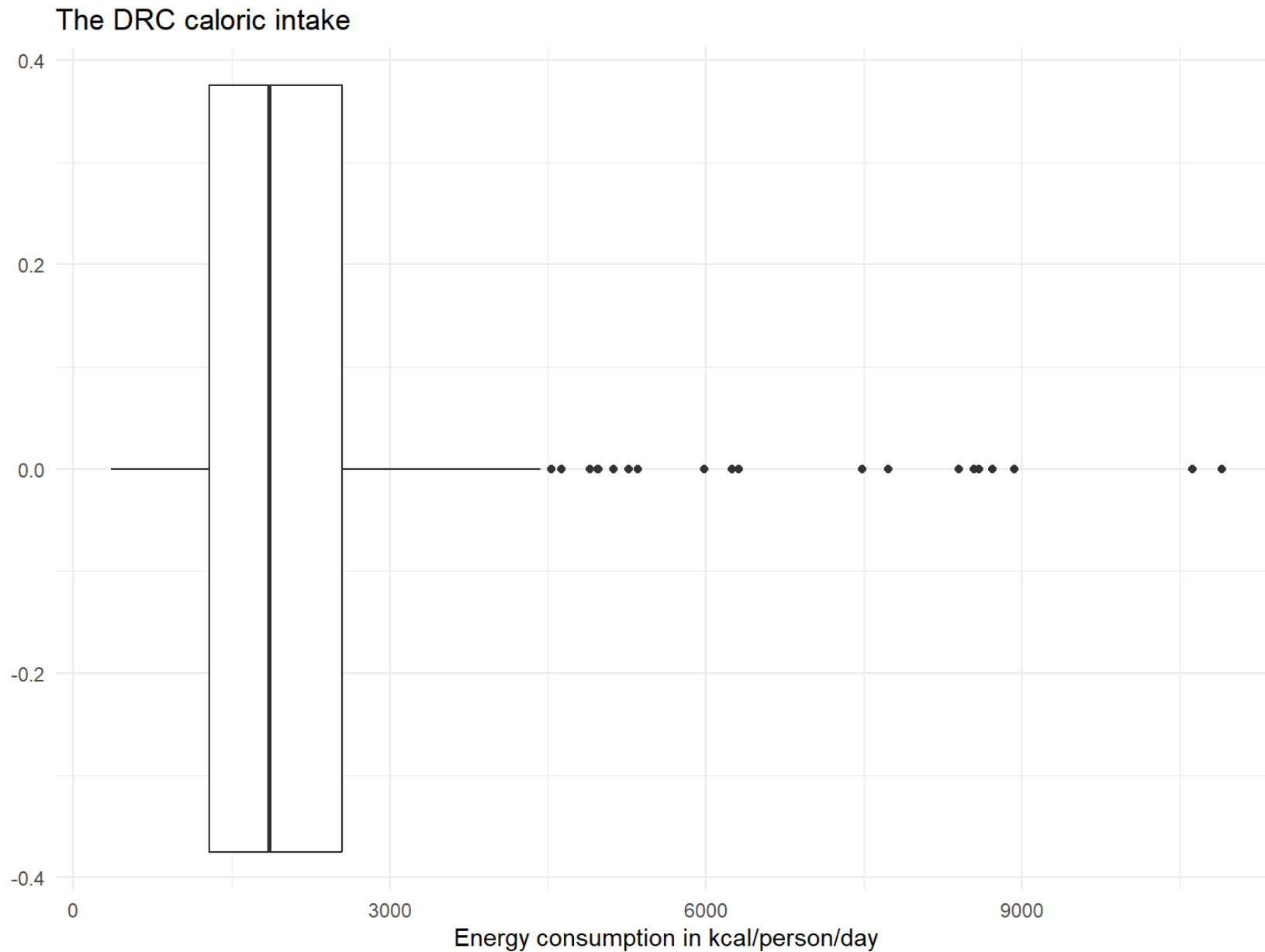


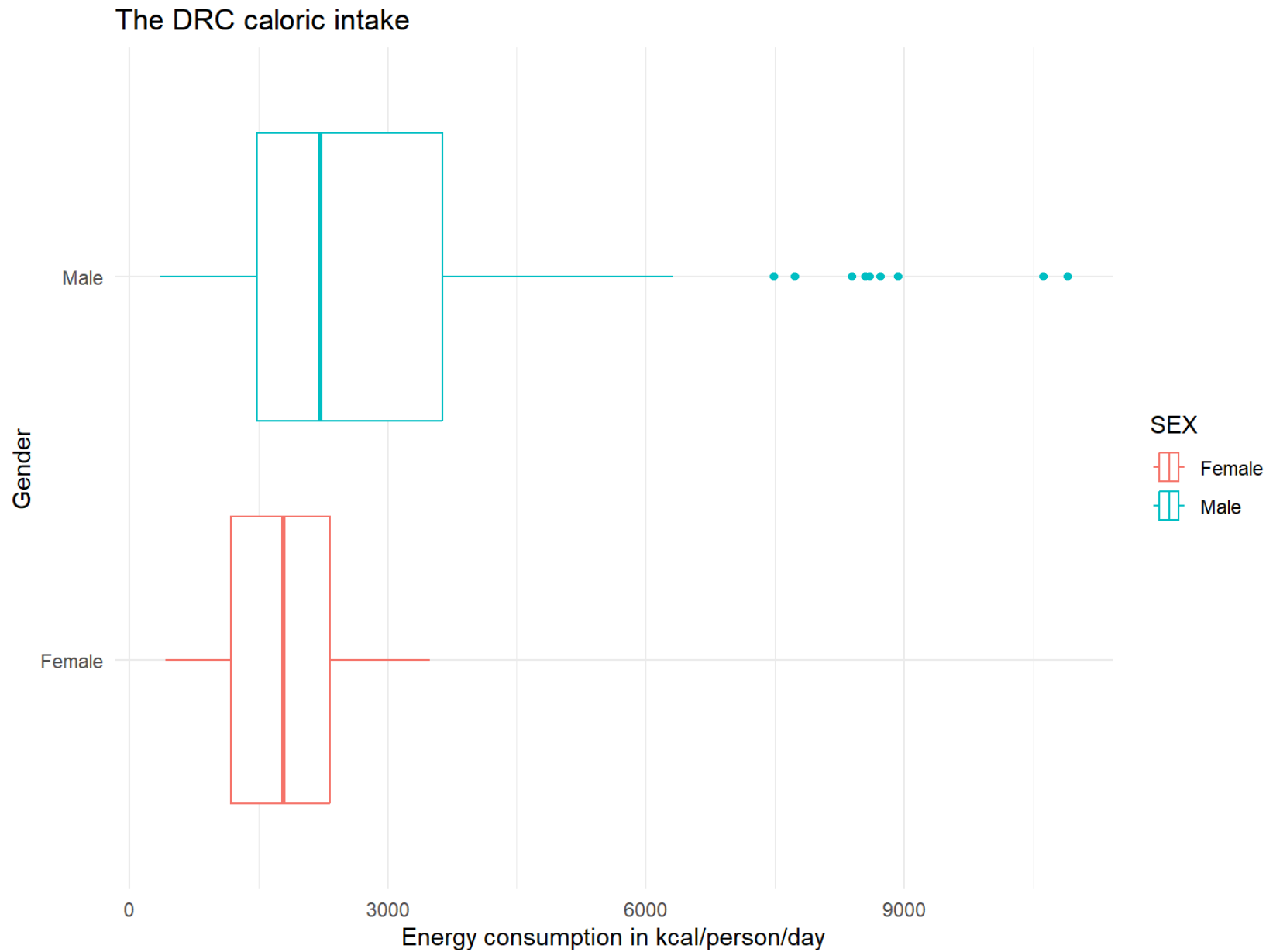
Image source

Single variable description with a Box Plot



- The central tendency;
- Where are the 50% of all observations?
- Dispersion of the values: **the Variance**;
- Min/Max;
- (Suspected) outliers;

Comparing distributions with Box Plots (2)



- Variance; Central tendency; (Suspected) outliers; Min/Max.

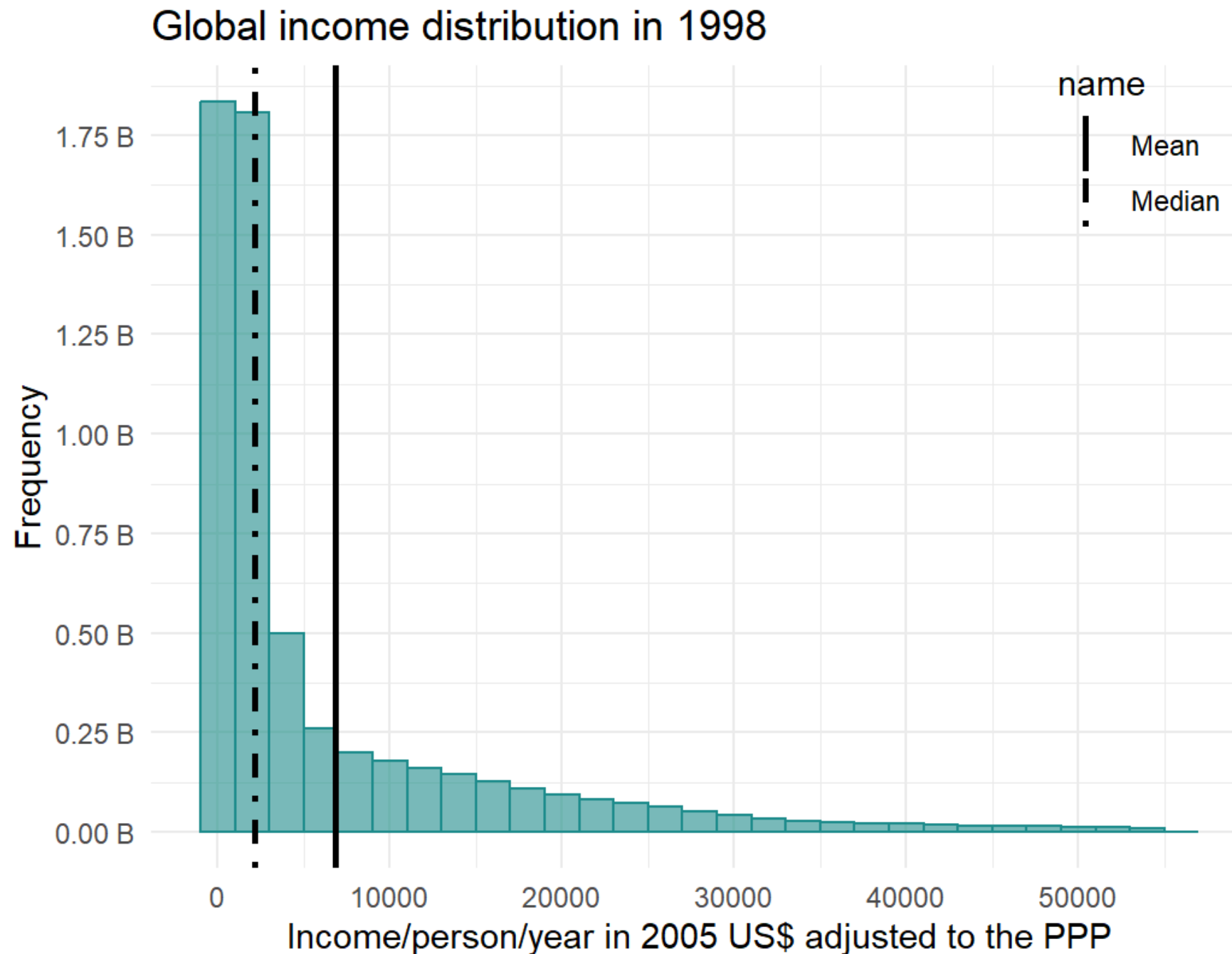
Global income distribution

The Data individual's **income** across the world.

- Combines many surveys from different countries
- Number of observations: 10+ million
- Observation units: **individuals**
- The population: **all people in the World**
- The variable: **income per person per year** in constant 2005 US\$ adjusted to the PPP ([Purchasing Power Parity](#)).
- Data sources: ([Lakner and Milanovic 2015](#)) at elibrary.worldbank.org, ([Roser 2013](#)) at OurWorldInData.org

See also: ([Wilke 2019](#)) [Ch. 8.2 Highly skewed distributions](#)

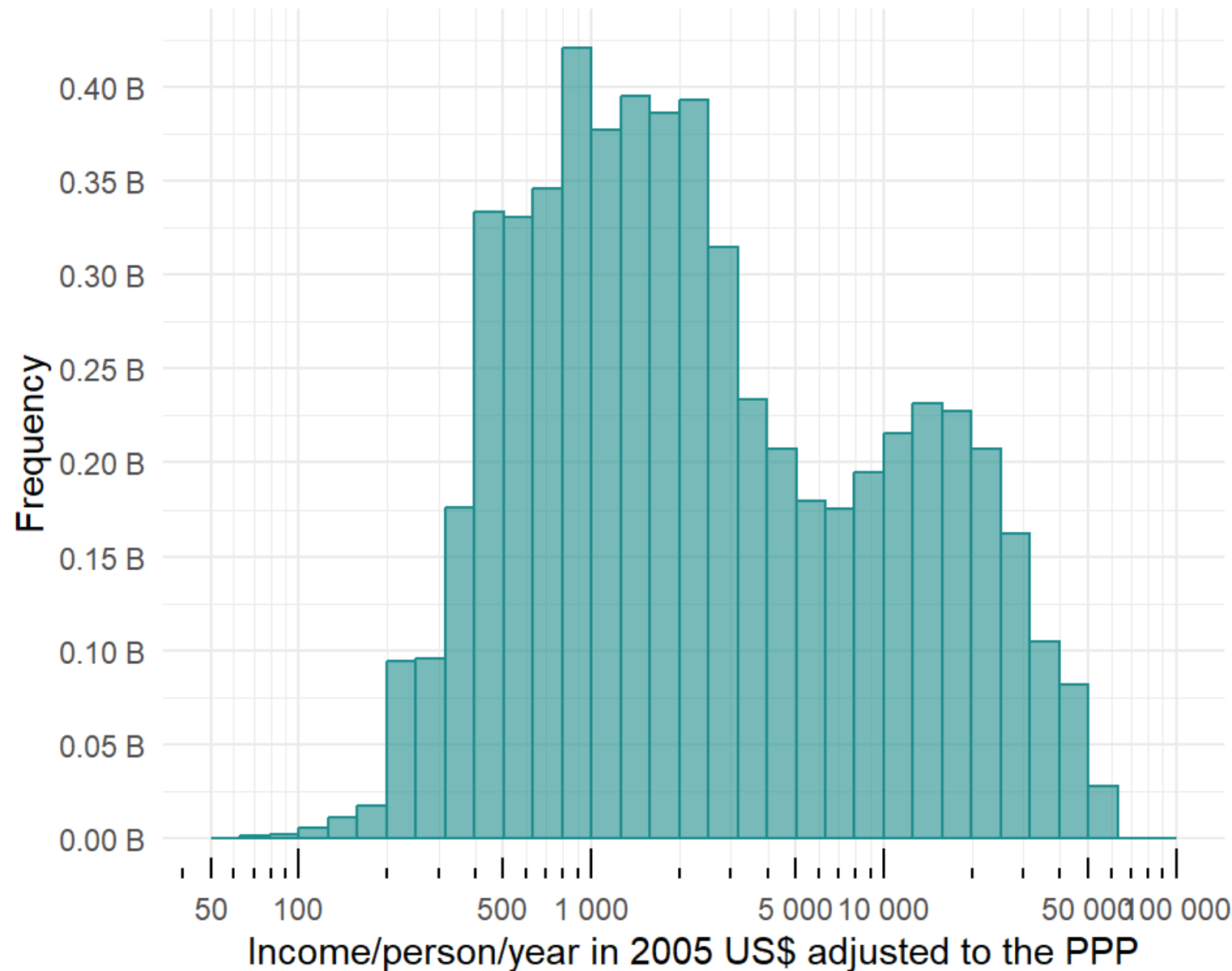
Income distribution



- How informative is this chart?
- Can we say anything about the distribution of income in the world?
- How can we improve the histogram?

Band width: Logarithm transformation

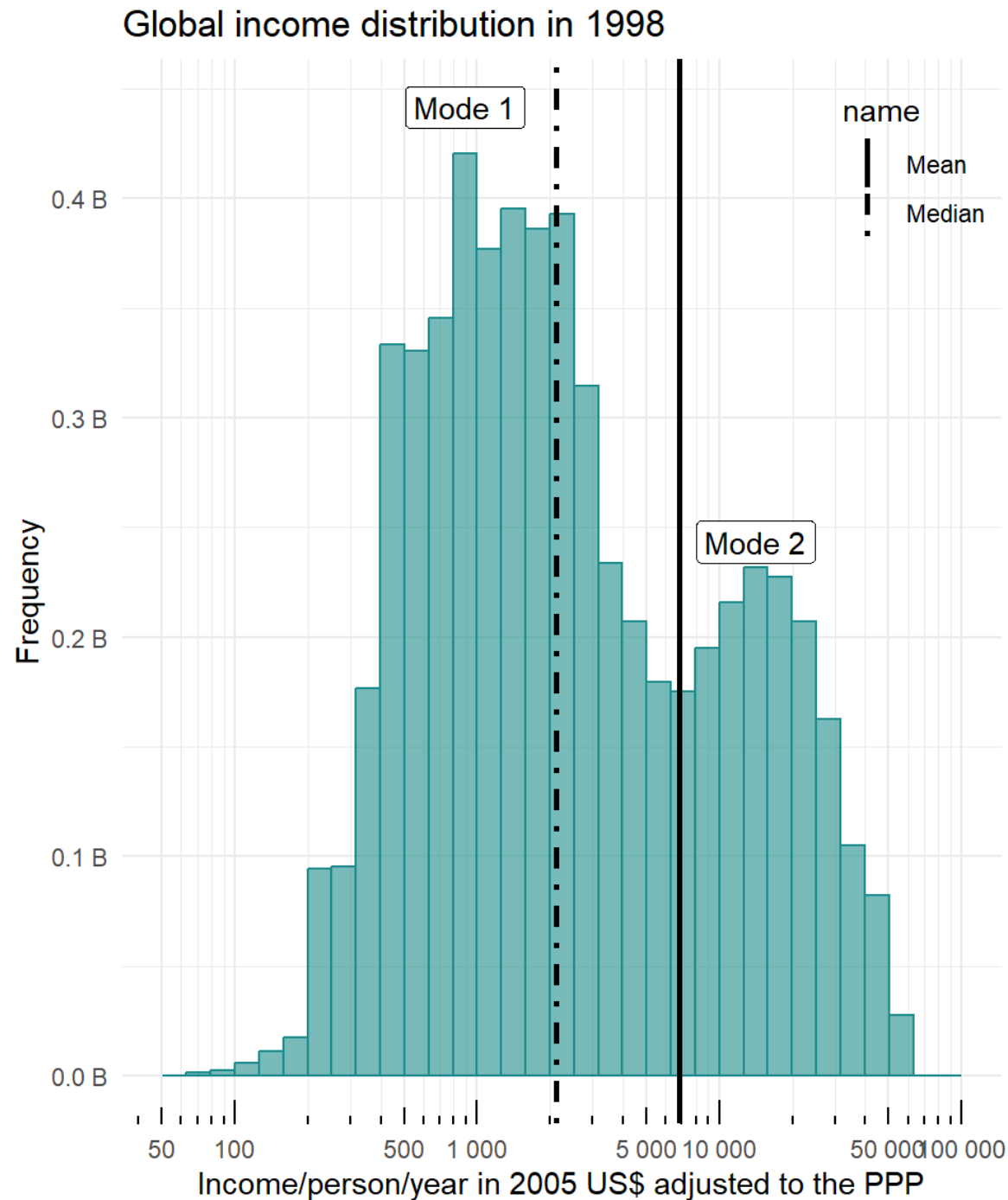
Global income distribution in 1998



The logarithm transformation?

- $\log_{10}$
- $\log_{10} 1 = 0$
- $\log_{10} 10 = 1$
- $\log_{10} 100 = 2$
- $\log_{10} 1000 = 3$
- $\log_{10} 10000 = 4$
- $\log_{10} 100000 = 5$

New insights (1)



What can we see now?

1. The mode:

- **Bi-modal distribution.**
- What is “bi-modal”?
- What causes two spikes?

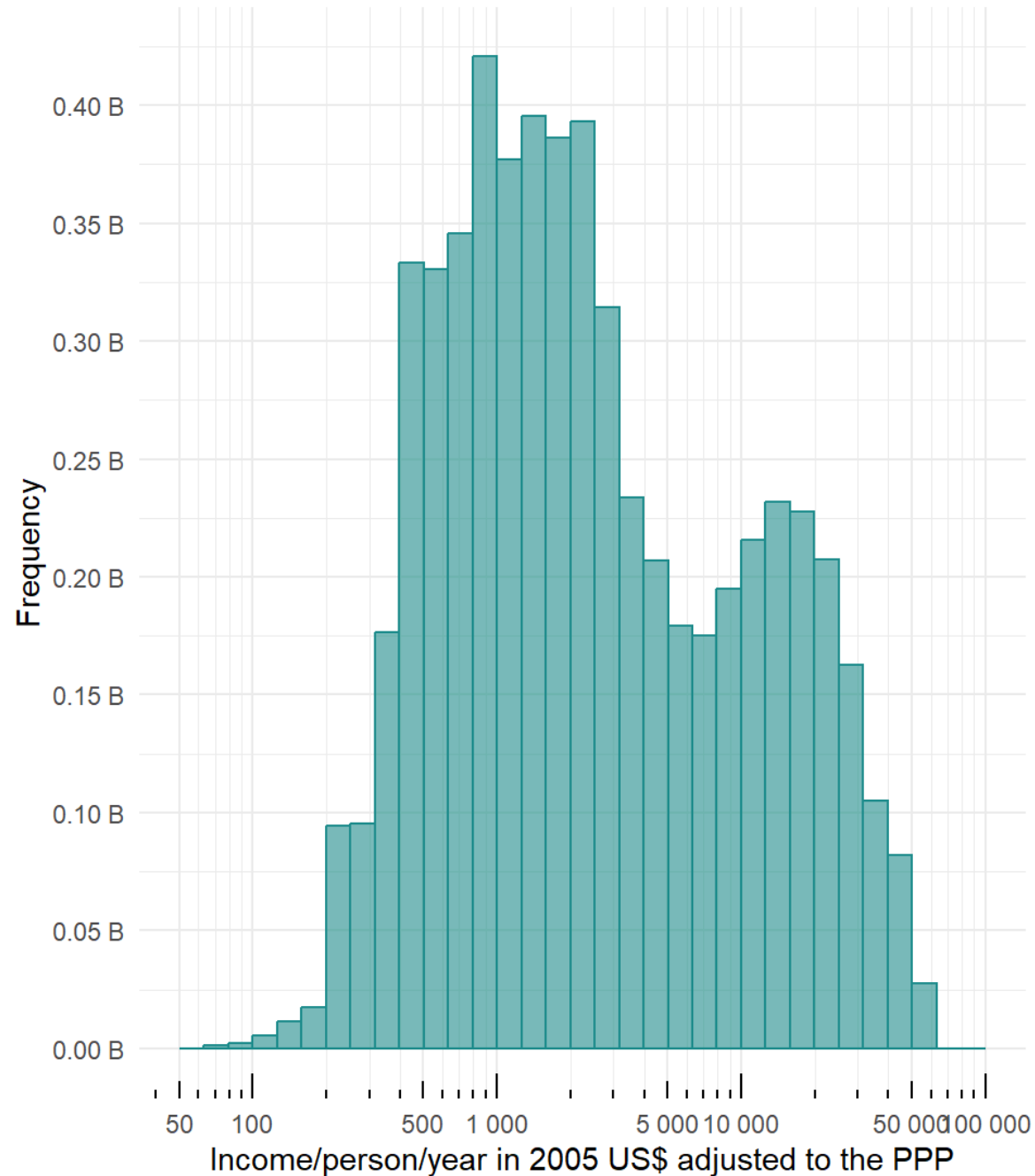
2. The mean and the median

- They are more evident after transformation.

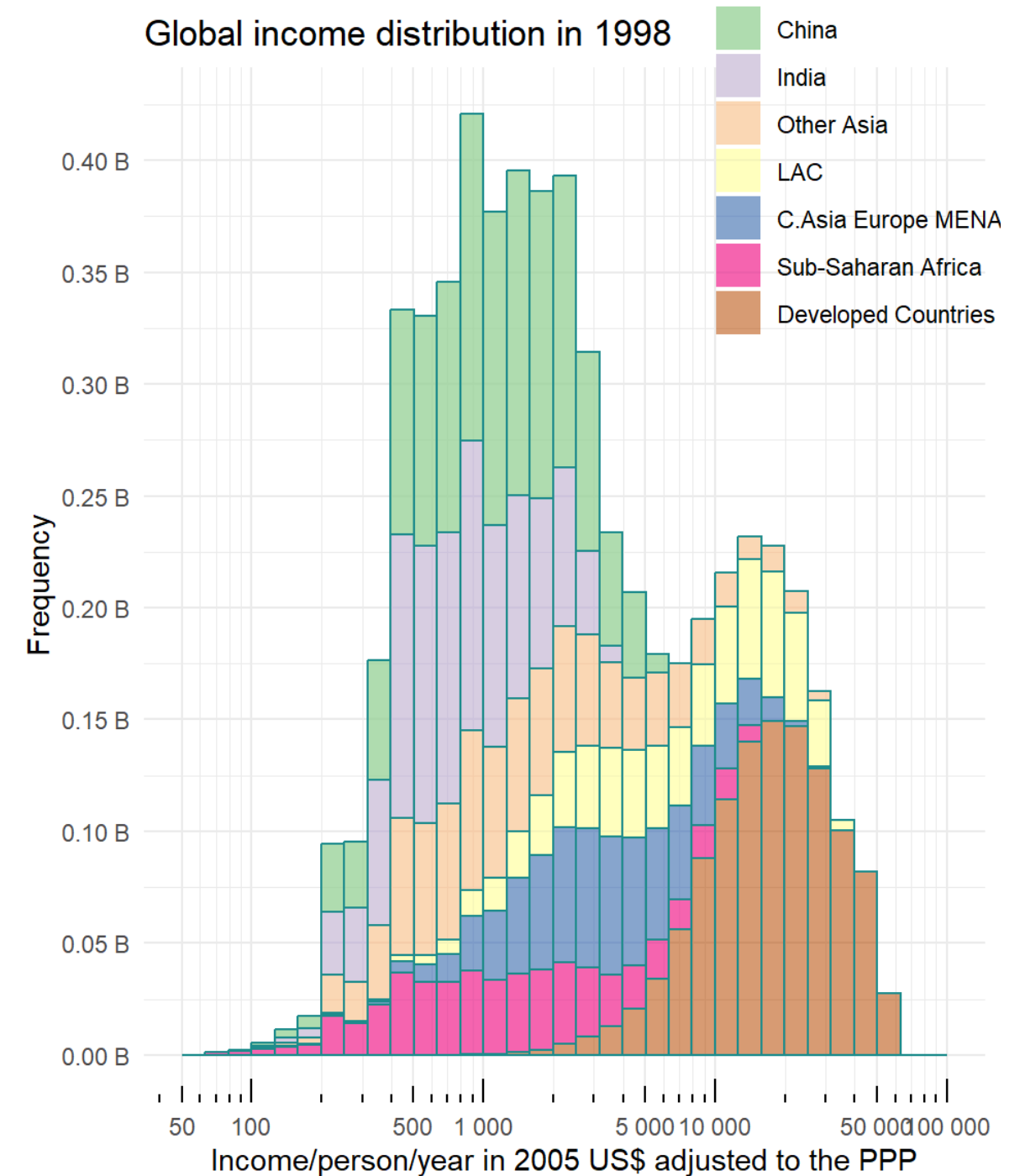
Why are there two modes?

New insights (2): data complexity

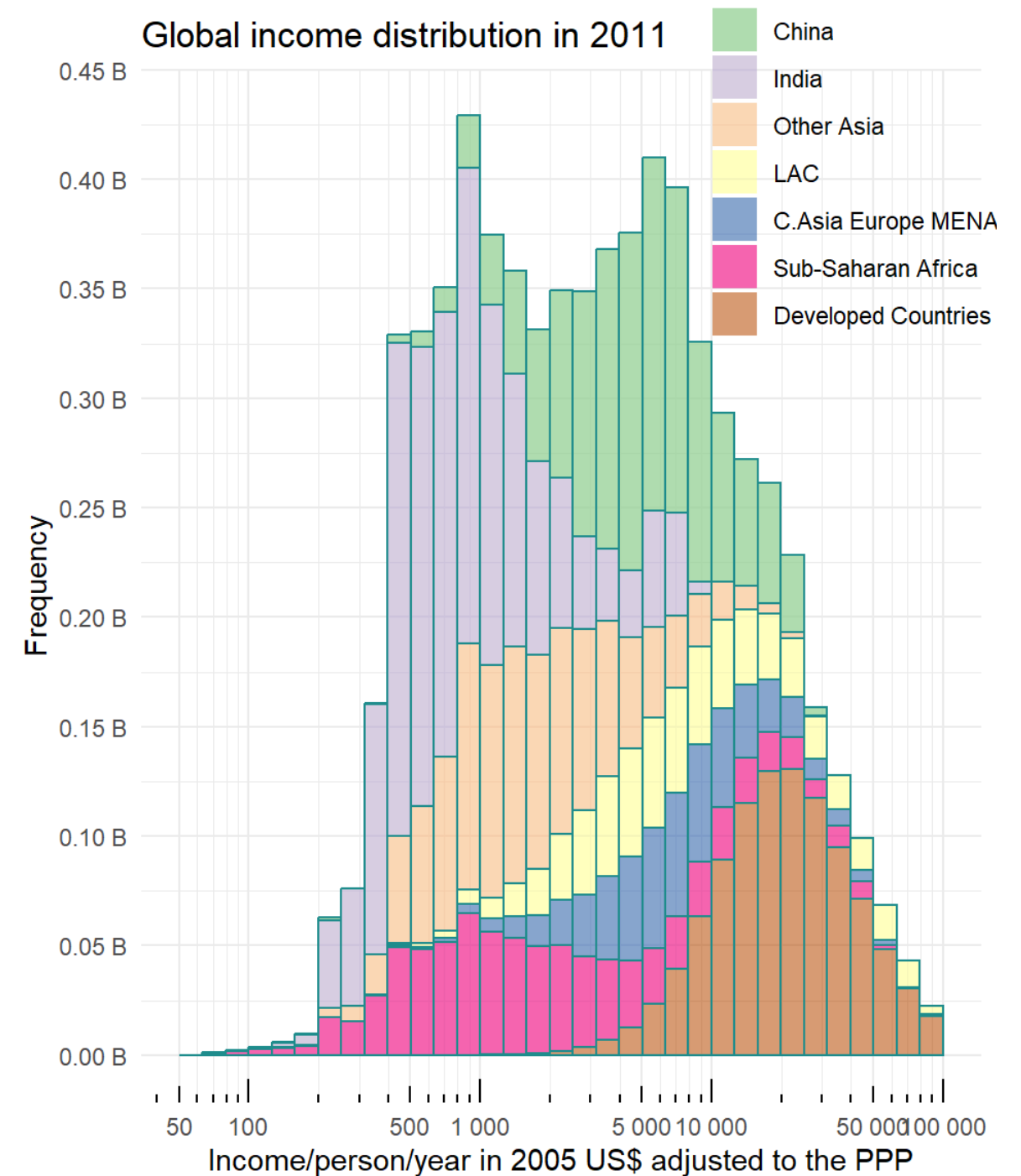
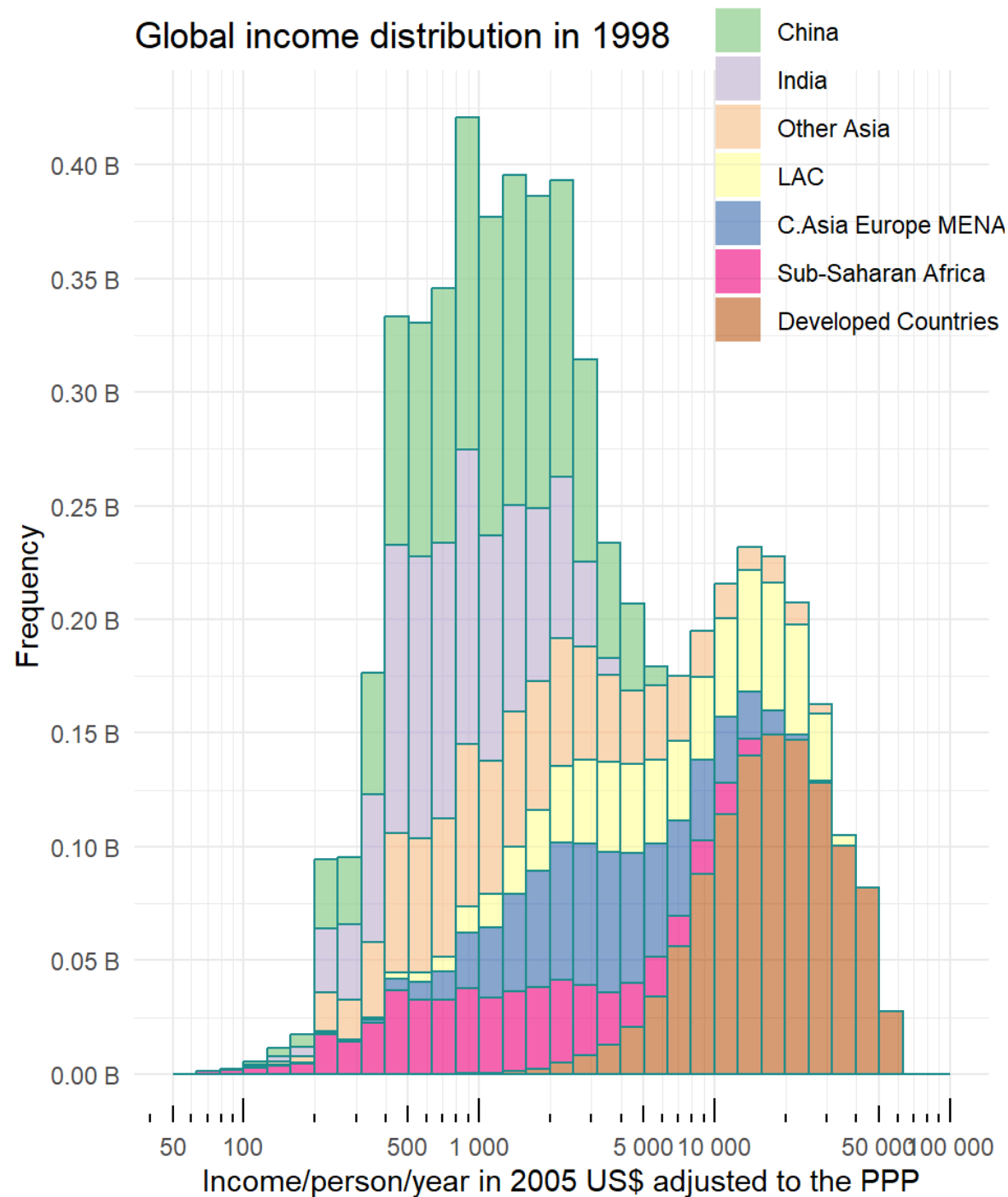
Global income distribution in 1998



Global income distribution in 1998

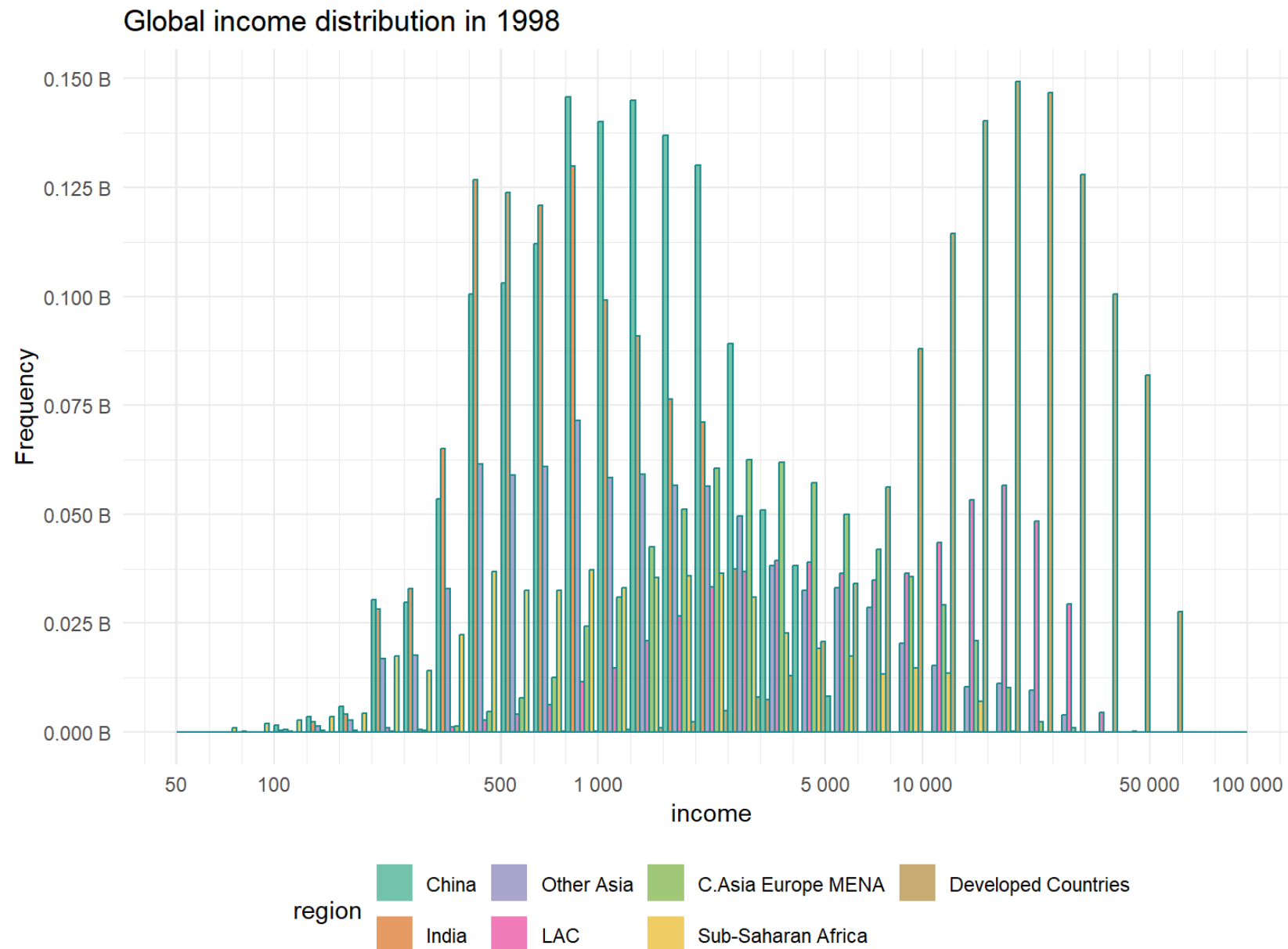


New insights (3): change over time



Comparing distributions

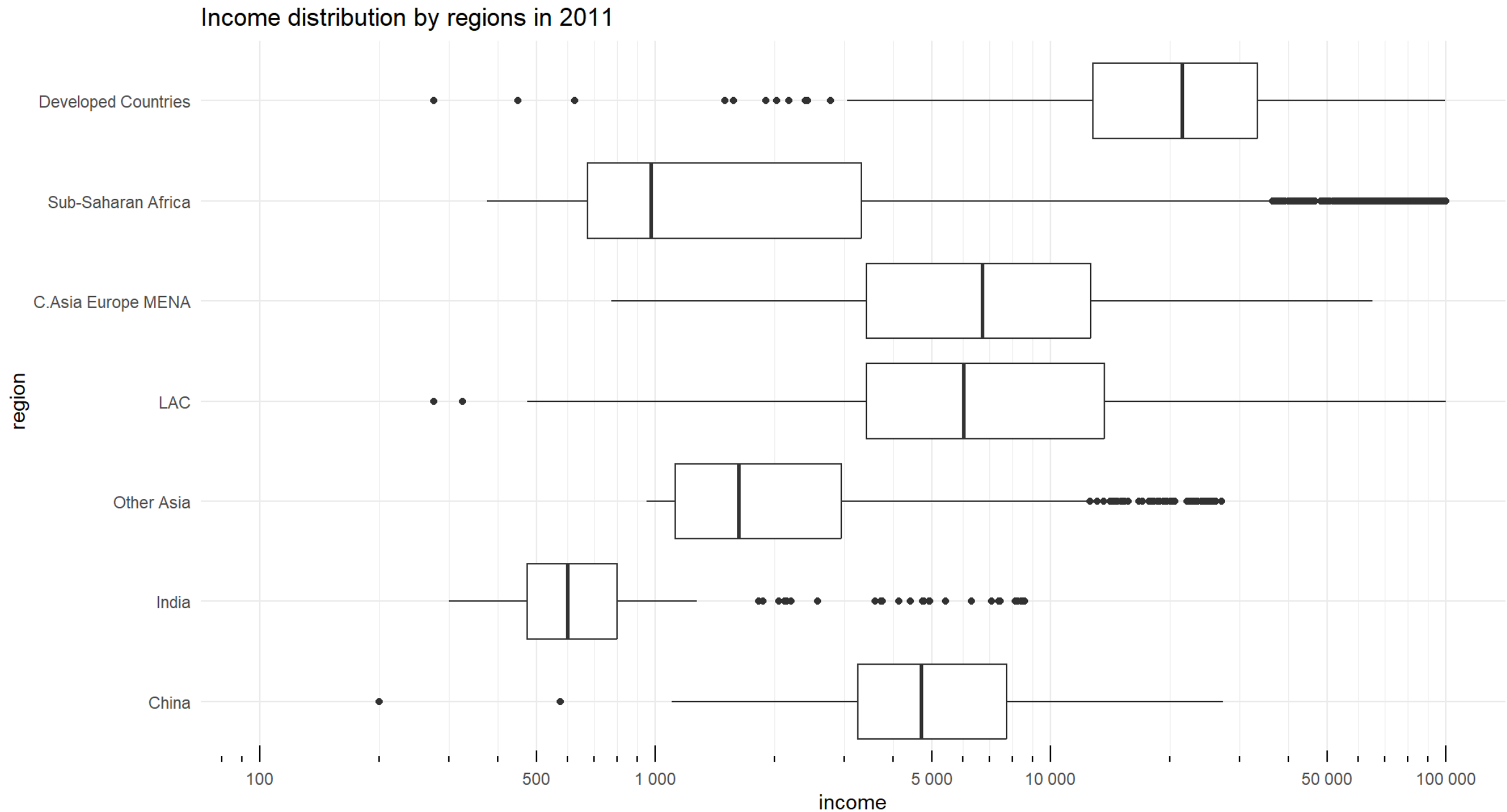
How to compare distributions of income between regions?



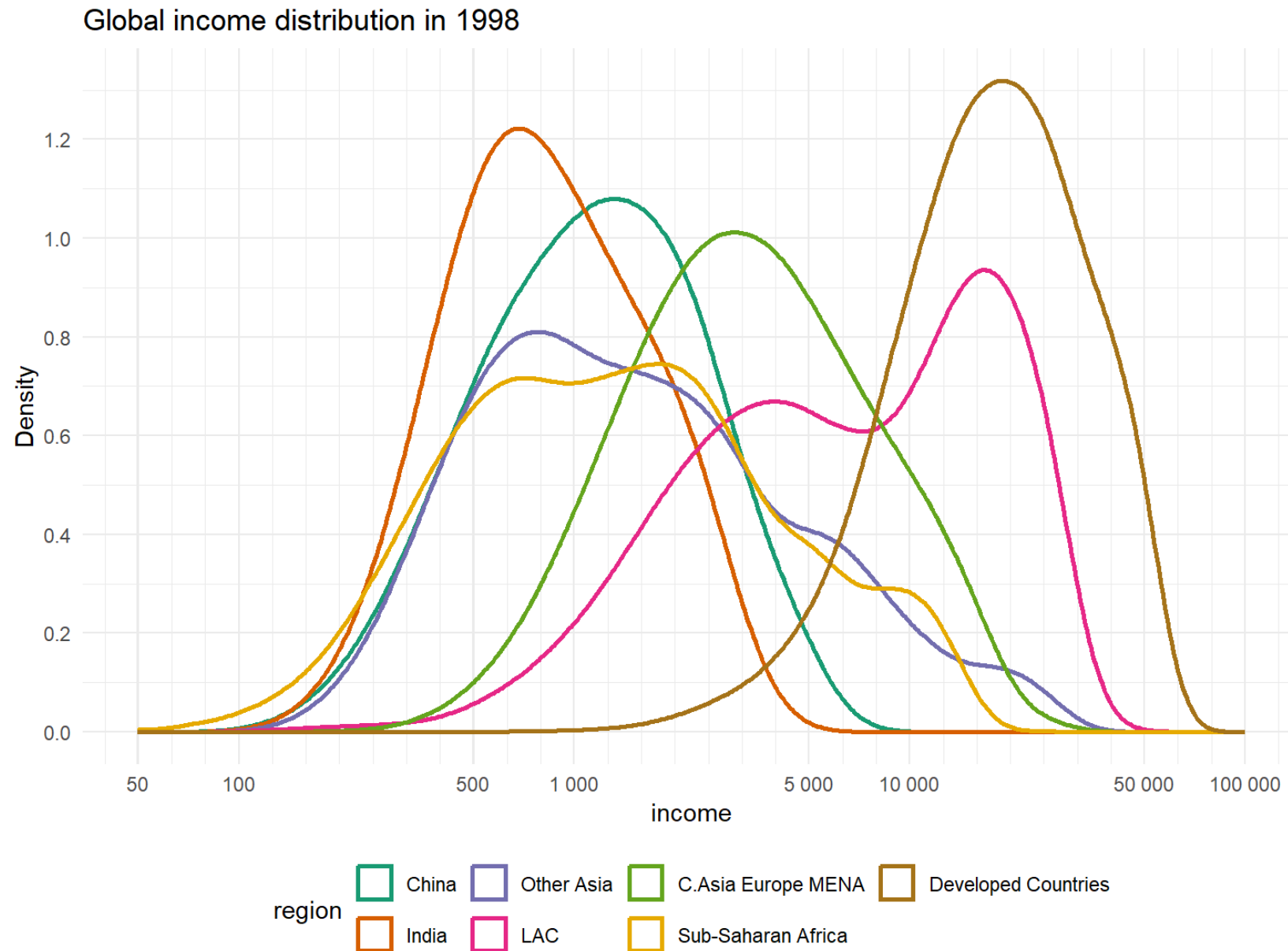
We can use histograms, but is it informative?

- Box plots?
- Density curves?

Box plots for comparing distributions



Density curves

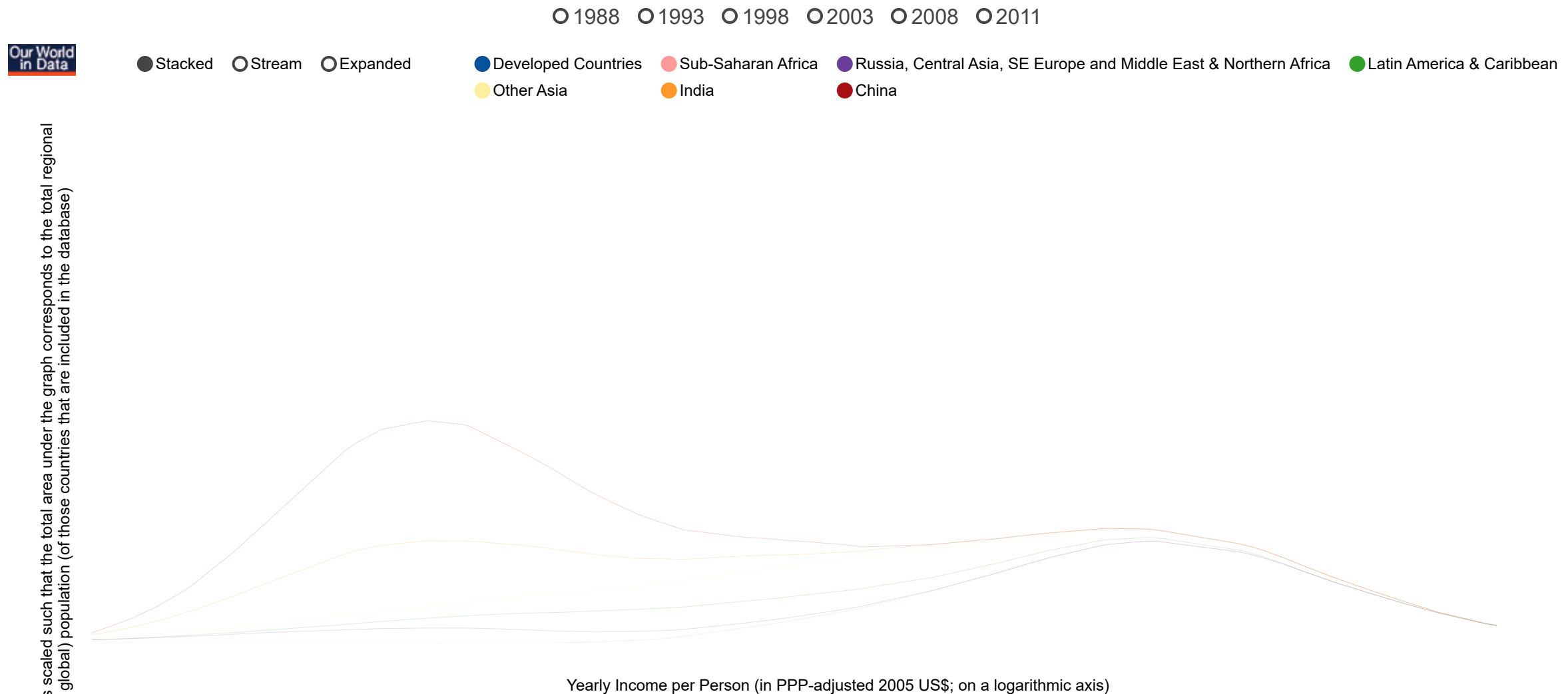


- This is a density curve.
- Area under the curve is 1.
 - Density plot allows to compare the distributions.

What do we learn from this plot?

- Density “location”.
- Distribution kurtosis.

At home: Explore all the aspects of income inequality at [OurWorldInData.org](https://ourworldindata.org)

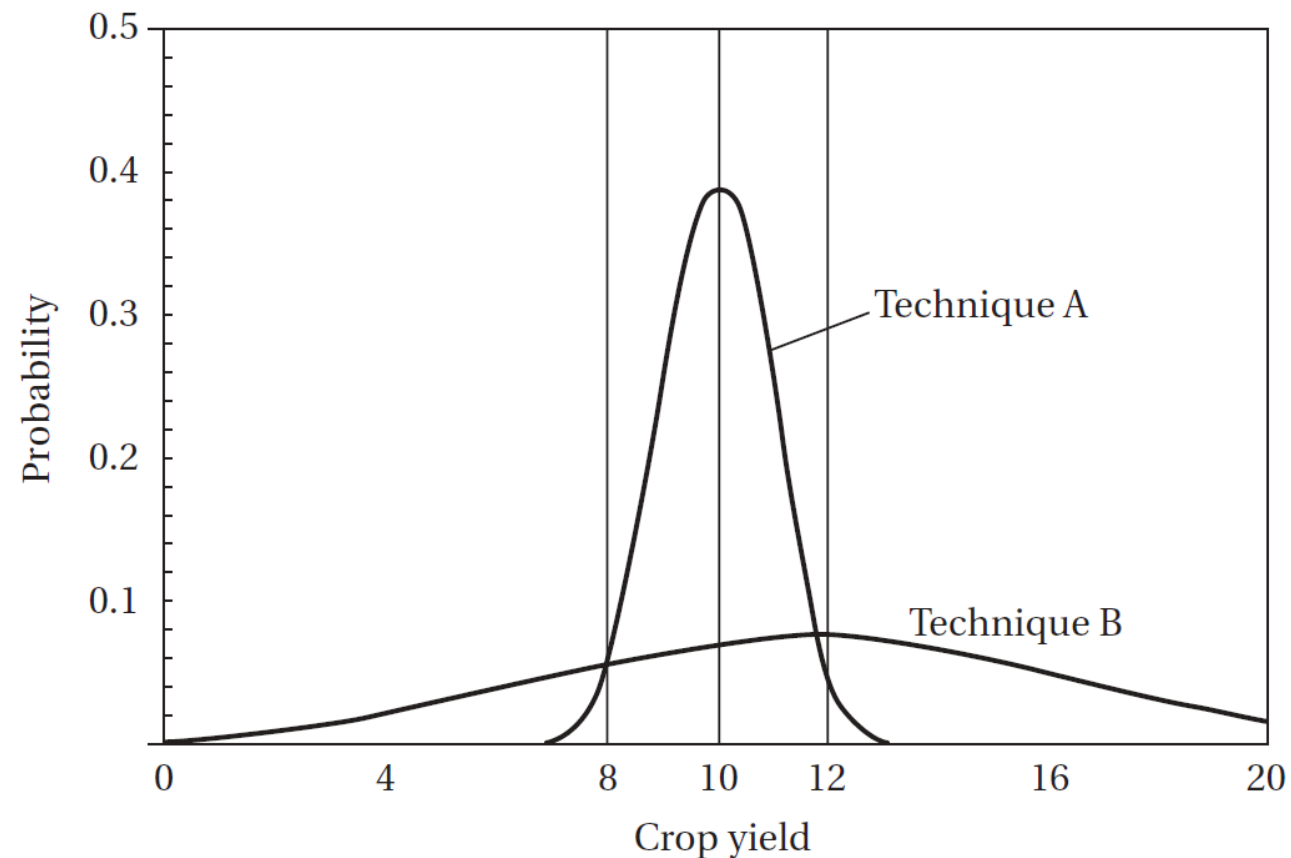


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Citation to the paper: [Global Income Distribution: From the Fall of the Berlin Wall to the Great Recession](#), Christoph Lakner and Branko Milanovic, World Bank Economic Review, Advance Access published August 12, 2015.
Data Visualization by Max Roser and Zdenek Hynek

At home: Exam problems on Density (Examples)

Crop yields and technology

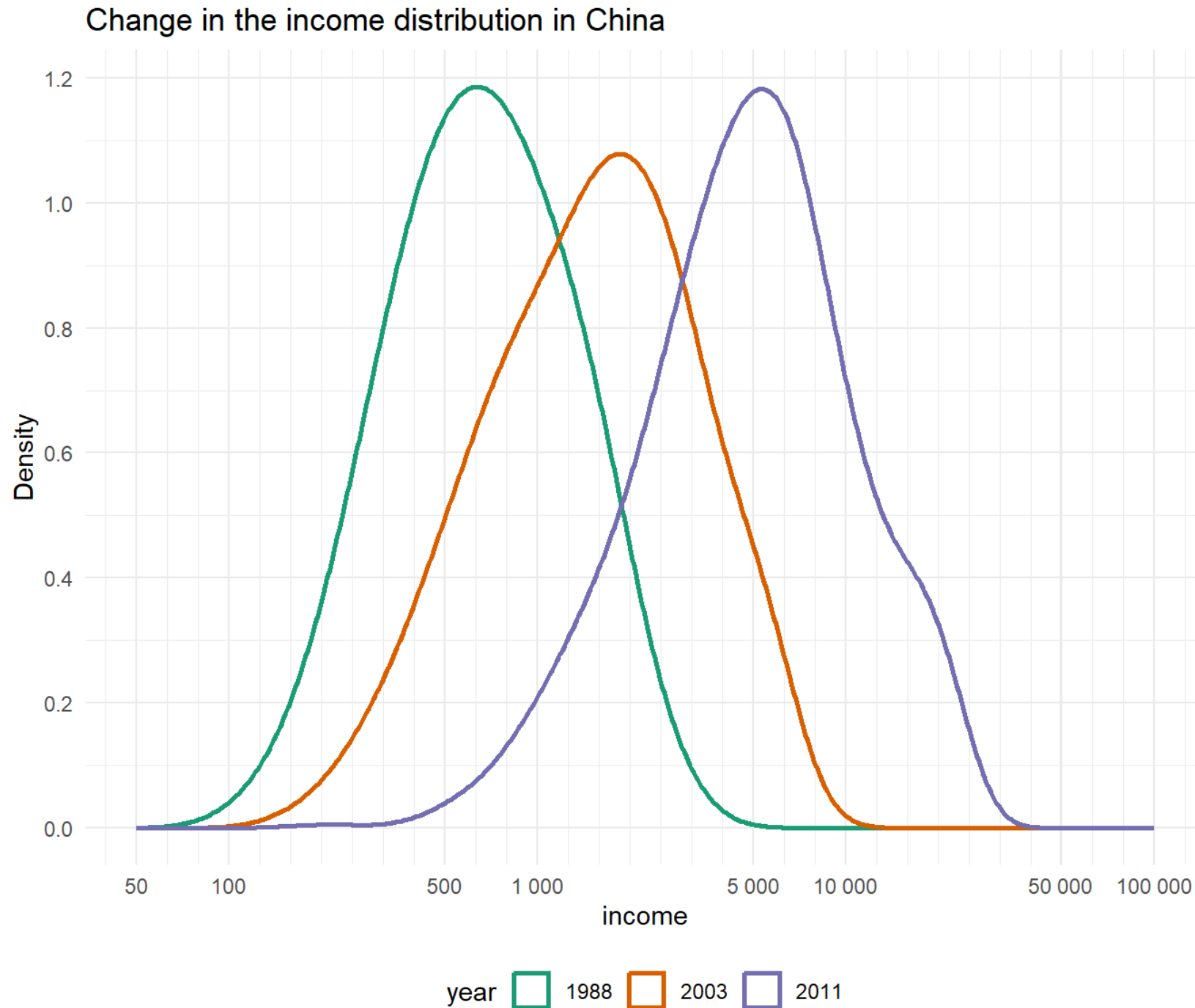
FIGURE 9.6 Crop Yield Probability Densities of Two Different Farming Techniques



- What does this plot show?
- What can we conclude about the distribution in the **Technique B**?
- How are the **Technique A** and the **Technique B** comparable?
 - Centers and variances;

Source: (Todaro and Smith 2014)

Income distribution change



What do we learn from this plot?

- Income growth over time.
- Emergence of the richer groups of population.

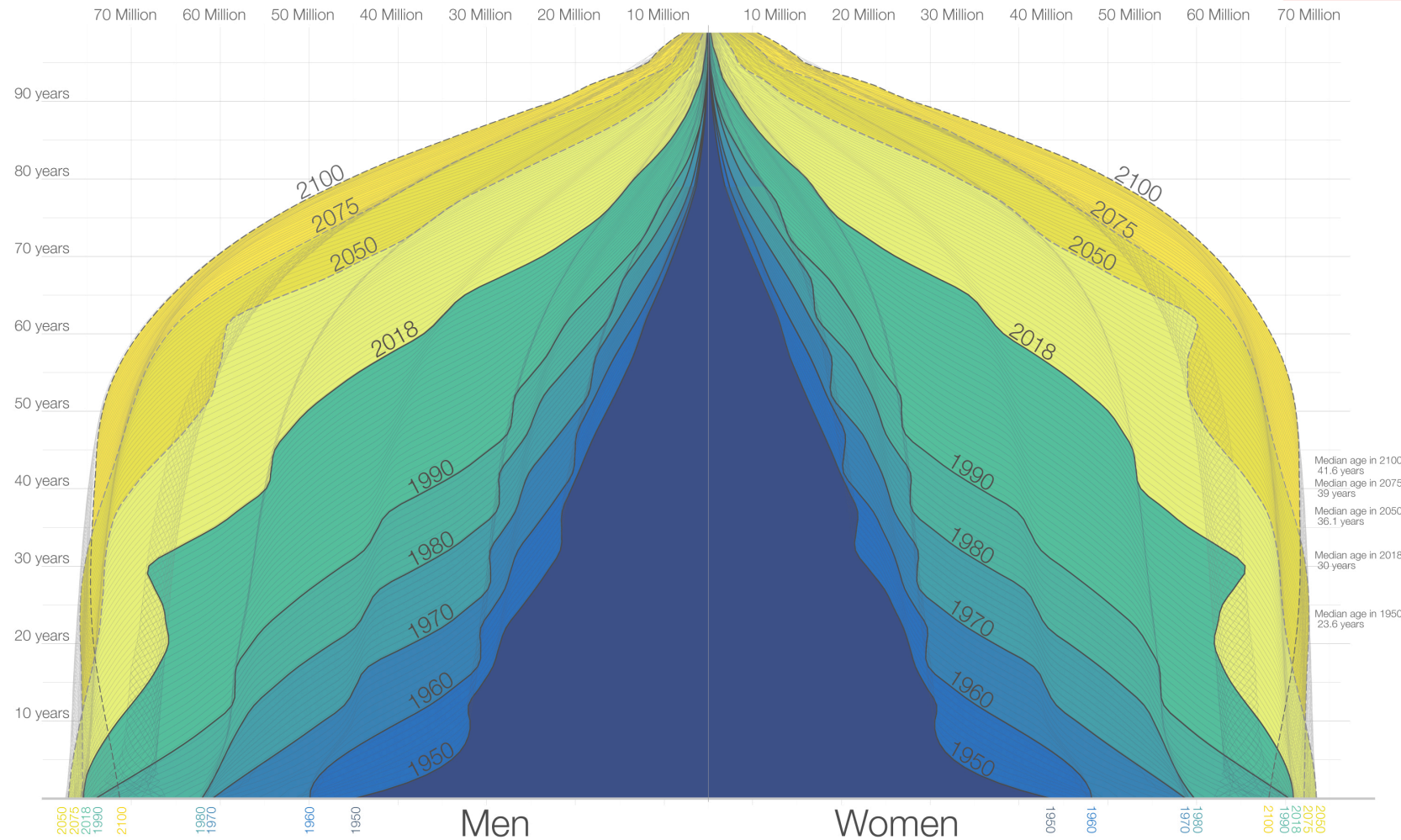
Data source: ([Roser 2013](#)). Published online at [OurWorldInData.org](#)

Population distribution by age

The Demography of the World Population from 1950 to 2100

Shown is the age distribution of the world population – by sex – from 1950 to 2018 and the *UN Population Division's* projection until 2100.

Our World
in Data



Data source: United Nations Population Division – World Population Prospects 2017; Medium Variant.
The data visualization is available at [OurWorldinData.org](https://ourworldindata.org), where you find more research on how the world is changing and why.

Licensed under CC-BY by the author Max Roser.

What do we learn from this plot?

- Does it tell us anything about the size of the population?
- What age groups will drive global population in 2050 compared to 2018.

Source: [World Population Growth](https://ourworldindata.org)

Takeaways checklist

Homework

Read the paper below **before week 4**: ([Kuchenbecker et al. 2017](#))

- Kuchenbecker, J., Reinbott, A., Mtimuni, B., Krawinkel, M. B., & Jordan, I. (2017). Nutrition education improves dietary diversity of children 6-23 months at community-level: Results from a cluster randomized controlled trial in Malawi. PLOS ONE, 12, e0175216.

Key concepts covered in the lecture

Univariate data

The Histogram:

The Density plot:

- Density vs frequency;
- How to make a density plot;

The Central tendency:

- Median and mode.
- Mean, mean of a dummy vs continuous variable;
- Extreme and influential observations and how to deal with it.
- Weighted average.

Box plot:

- Key elements and messages and how to construct it.
- Comparing distributions with the box plot.

References

- Kuchenbecker, Judith, Anika Reinbott, Beatrice Mtimuni, Michael B. Krawinkel, and Irmgard Jordan. 2017. “Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi.” Edited by Jacobus P. van Wouwe. *PLOS ONE* 12 (4): e0175216. <https://doi.org/10.1371/journal.pone.0175216>.
- Lakner, Christoph, and Branko Milanovic. 2015. “Global Income Distribution from the Fall of the Berlin Wall to the Great Recession.” *Revista de Economía Institucional* 17 (32): 71–128.
- Roser, Max. 2013. “Global Economic Inequality.” *Our World in Data*.
- Todaro, Michael, and Stephen C. Smith. 2014. *Economic Development*. 11th ed. Pearson.
- Wilke, Claus O. 2019. *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O’Reilly Media.